Strategy Proposal: Insider Signal Earnings Momentum Trader (ISEM)

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1. Executive Summary

This proposal outlines a new short-term swing trading algorithm — the **Insider Signal Earnings Momentum Trader (ISEM)** — built on the premise that corporate insiders and members of Congress, when purchasing stock in the period between earnings reports, often possess informational advantages about the company's next reported results. By cross-referencing mandatory public disclosures of such purchases against the upcoming earnings calendar, the algorithm identifies stocks likely to rise on their next earnings report. ISEM takes a single long position ahead of each qualifying earnings event, holds the position through the market's processing of that report, and exits at close the following day. All trades are bracket orders with a stop-loss. If no high-conviction candidate exists on a given day, the algorithm holds cash.

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2. Investment Thesis

When a company executive purchases a meaningful amount of their own company's stock in the weeks leading up to an earnings report — particularly as that report approaches — it is reasonable to infer that the executive has confidence in results that have not yet been made public. Public companies trading on major U.S. exchanges (NYSE, NASDAQ, etc.) are required by the SEC to disclose such purchases via **Form 4**, typically within two business days of the transaction. Members of Congress are separately required to disclose their stock trades within 45 days of a transaction under the STOCK Act.

If a company beats earnings expectations, its stock price typically rises on or immediately after the announcement. By systematically identifying stocks where insiders and Congress members have been purchasing shares during the inter-earnings window — especially with increasing frequency as the next earnings date approaches — the algorithm positions itself to capture that post-earnings upside, while a strict stop-loss limits downside exposure.

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3. Insider Universe

The algorithm will monitor purchases made by the following categories of individuals for each company with an upcoming earnings event:

Corporate Insiders (via SEC Form 4)

- C-suite executives (CEO, CFO, COO, CTO, President, etc.)
- Board of directors members
- Officers designated as "reporting persons" by the SEC
- Beneficial owners of 10% or more of outstanding shares

Government Insiders (via STOCK Act Disclosures)

- U.S. House of Representatives members
- U.S. Senate members

Transaction Type Filter: The algorithm should only track **open market purchases** (Form 4 transaction code "P"). It should exclude option exercises, awards, gifts, and planned purchases made under pre-scheduled 10b5-1 trading plans, as these do not reflect real-time conviction. A flag for 10b5-1 plan purchases is included in Form 4 filings and should be used to filter these out.

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4. Data Sources

All recommended sources below are free of charge.

4.1 Earnings Calendar

- **Primary:** Yahoo Finance via the `yfinance` Python library — provides the earnings calendar, company tickers, and report timing (before market open, during market hours, or after market close)
- **Secondary:** Nasdaq Earnings Calendar — useful for cross-referencing and validating report timing

4.2 Corporate Insider Transactions (Form 4)

- **Source:** SEC EDGAR Full-Text Search API and the SEC EDGAR Bulk Data
- Form 4 filings must be submitted within 2 business days of the transaction, so this data is near real-time
- Each filing specifies: the filer's role, transaction date, shares purchased, price per share, and whether the trade was part of a 10b5-1 plan
- The `sec-edgar-downloader` Python package simplifies bulk retrieval

4.3 Congressional Disclosures

- **House:** House Stock Watcher — free JSON API for all House member disclosures
- **Senate:** Senate Stock Watcher — free JSON API for all Senate member disclosures
- **Important Lag Note:** Congressional members have up to 45 days to file disclosures. The algorithm must use the **transaction date** listed in the disclosure, not the disclosure date, when applying the timing windows described in Section 5. A purchase disclosed today may have occurred 6 weeks ago. This is a critical distinction.

4.4 Stock Price Data

- **Source:** Yahoo Finance via yfinance — free, no API key required
- Provides historical OHLCV data for backtesting and real-time quotes for trade execution
- **Backup:** Alpha Vantage (free tier: 25 requests/day) for cross-validation

4.5 Prior Earnings Dates (for Window Calculation)

- Also available via `yfinance` — the `Ticker.earnings_dates` property returns historical earnings dates, which are needed to define the start of each inter-earnings signal window

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5. Signal Timing Framework

The period between consecutive earnings reports is divided into distinct timing zones. Insider purchases carry different signal weight depending on when they occurred within the inter-earnings cycle.

5.1 Post-Earnings Exclusion Window (Days 1–14 after prior earnings)

- Duration: Approximately 14 calendar days (10 trading days) following the prior earnings report date
- **Rationale:** Purchases immediately after an earnings report are most likely a reaction to newly revealed results — good results that are now public knowledge — rather than forward-looking signals about the *next* earnings period. These purchases do not help predict future, unreleased performance.
- **Treatment:** Excluded entirely from the scoring algorithm.
- **Note:** This window is a recommended starting point. The exact cutoff (10, 14, or 21 days) should be validated and tuned during backtesting, and the architecture should make this parameter easily adjustable.

5.2 Core Signal Window (Day 15 after prior earnings through Day 28 before next earnings)

- This is the primary zone of interest. Purchases here are most likely driven by genuine confidence in current business performance during the inter-earnings period.
- Most publicly traded companies enforce formal trading blackout periods beginning approximately 30 days before their earnings announcement. Purchases within the Core Signal Window occur during the legally permissible open trading window and carry strong credibility.
- **Treatment:** Full scoring weight applied. See Section 6.

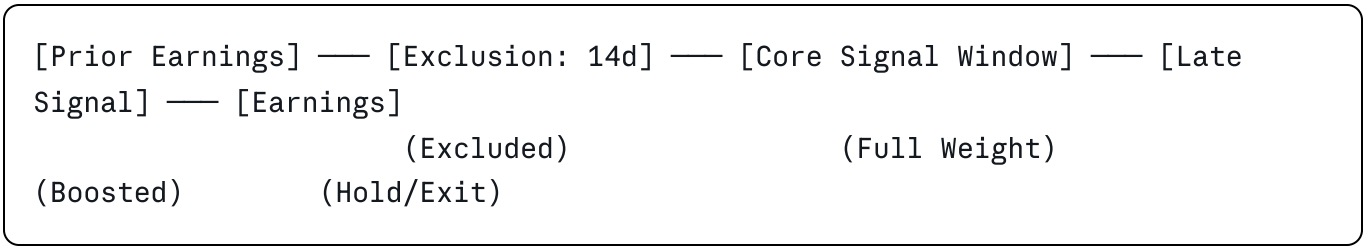
5.3 Late Signal Window (Day 28 through Day 3 before next earnings)

- As the next earnings date approaches, insiders are increasingly likely to be constrained by their company's blackout policy. Any purchases in this window despite a typical blackout would be unusual and potentially indicate extremely high conviction (though they also raise compliance questions).
- In practice, some companies have shorter or no formal blackout periods, so late-window purchases may simply reflect normal trading.
- **Treatment:** Purchases in this window receive a **signal boost** — they are weighted more heavily than Core Signal Window purchases, as proximity to the earnings date with continued buying suggests heightened confidence. The exact multiplier should be calibrated during backtesting.

5.4 Final 3 Days Before Earnings (Day 3 to Day 0)

- This is the algorithm's own entry zone. No new insider signal analysis is applied here; the algorithm is executing, not still evaluating.

5.5 Diagram of Signal Windows



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6. Insider Signal Scoring

For each company with an upcoming earnings event, the algorithm generates a composite signal score based on all insider and congressional purchases within the relevant timing

windows. Only companies that meet a minimum score threshold are considered for trade entry. The minimum threshold should be determined through backtesting.

6.1 Scoring Dimensions

A. Insider Count (max 25 points)

- Each unique insider or Congress member who made a qualifying purchase: **+5 points each**, up to a maximum of 5 distinct buyers

B. Role Weighting (max 20 points)

- At least one C-suite executive among buyers: **+10 points**
- At least one board member or beneficial owner among buyers: **+5 points**
- At least one Congress member among buyers: **+5 points** (equal weight to corporate insider per role tier)

C. Timing Quality (max 20 points)

- Core Signal Window purchases present: **+10 points**
- Late Signal Window purchases present: **+20 points** (replaces Core Window points if both windows have purchases — Late Window is the stronger signal)

D. Velocity Score (max 20 points)

- Multiple purchase events (across time) by any combination of insiders: **+10 points**
- Increasing purchase dollar amounts across the inter-earnings period (trend upward): **+10 points**
- Velocity is calculated by dividing the inter-earnings window into thirds and comparing purchase volume in each third

E. Congressional Lag Adjustment (up to +5 points)

- If a congressional purchase was disclosed very recently but the actual transaction date (per the filing) falls squarely in the Late Signal Window, add **+5 points** to reflect the recency of the underlying information

Total Maximum Score: 90 points

6.2 Score Interpretation (Suggested Starting Thresholds)

Score	Interpretation	Action
0 – 29	Weak or no signal	Hold cash

Score	Interpretation	Action
30 – 49	Moderate signal	Monitor; do not trade
50 – 69	Strong signal	Eligible for trade entry
70 – 90	Very strong signal	Priority candidate

These thresholds are preliminary and should be validated against historical data during backtesting.

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7. Daily Algorithm Workflow

The algorithm runs the following process each trading day:

Step 1: Earnings Calendar Pull (Pre-Market)

- Query the earnings calendar for companies reporting earnings:
  - **Today** (if reporting after market close today)
  - **Tomorrow** (if reporting before market open or during market hours tomorrow)
- Filter to companies listed on NYSE or NASDAQ only
- Record each company's ticker, earnings report date, and report timing (BMO / Intraday / AMC)

Step 2: Insider Data Pull

- For each company identified in Step 1:
  - Retrieve their most recent prior earnings date
  - Pull all Form 4 open-market purchases filed since that prior earnings date
  - Apply the 10b5-1 filter (exclude pre-planned purchases)
  - Pull congressional disclosures for the same ticker; translate disclosure dates to transaction dates
  - Apply the Post-Earnings Exclusion Window (discard purchases within 14 days of prior earnings)

Step 3: Score Each Candidate

- Apply the scoring framework from Section 6 to each remaining company
- Discard any company scoring below the minimum threshold

Step 4: Rank and Select

- Rank all qualifying companies by their composite score (highest to lowest)
- If a tie exists, use dollar volume of insider purchases as a tiebreaker
- Select the **top-ranked company** as the trade candidate for the day

Step 5: Check Existing Position

- If the algorithm currently holds a position that has not yet reached its exit trigger, do not enter a new trade — wait for the current position to close first before evaluating the next candidate

Step 6: Execute or Hold Cash

- If a qualifying candidate exists and no current position is held: execute the bracket order (see Section 8)
- If no candidate meets the threshold: hold cash for the day. **Do not force a trade.**

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8. Trade Execution Rules

8.1 Position Sizing

- Each position uses **50% of total available capital**
- The remaining 50% is held in cash at all times
- This applies even when conviction is very high — no exceptions

8.2 Entry Timing

Earnings Report Timing	Entry Time
Before Market Open (BMO) — tomorrow	Before market close **today**
During Market Hours (Intraday)	Before market close **today**
After Market Close (AMC) — today	Before market close **today**

All entries target execution as close to market close as practical (e.g., within the final 15–30 minutes of the session) to capture as much of the intraday price discovery as possible while ensuring the position is established before the report.

8.3 Exit Timing

Earnings Report Timing	Exit Time
Before Market Open (BMO)	Market close **on the day of the report**
During Market Hours (Intraday)	Market close **on the day of the report**
After Market Close (AMC)	Market close **on the next trading day**

The algorithm does not attempt to sell into the post-earnings spike in real-time. Selling at close allows the market a full session to process the earnings reaction before the position is liquidated.

8.4 Weekend Holding

- Positions may occasionally be held over weekends. This occurs when a company reports earnings after close on a Friday; the exit is at close the following Monday.
- The stop-loss bracket remains active, but gap risk over the weekend should be understood as an inherent characteristic of this strategy (see Section 10).

8.5 Bracket Order Structure

Every trade is entered as a **bracket order** consisting of:

1. **Market/Limit Entry Order** — executed near market close on the entry day (see Section 8.2)
2. **Stop-Loss Order** — a sell stop set at **4% below the entry price**
    - This is a hard protective floor. If the stock falls 4% from entry before the scheduled exit, the position is closed immediately
    - The 4% level is chosen to allow for normal overnight volatility while protecting against meaningful adverse moves. This figure should be reviewed and adjusted based on backtesting results
3. **Primary Exit Order** — a market-on-close (MOC) sell order for the designated exit date (see Section 8.3)

The entry and primary exit are linked; the stop-loss is a standing resting order that cancels automatically if the primary exit executes first. All three legs are submitted together.

Gap Risk Acknowledgment: In cases where a stock opens significantly below the stop-loss trigger (e.g., due to a pre-market earnings miss), the stop order may execute at a price meaningfully worse than 4% below entry. This is a known and accepted risk of overnight holding and cannot be fully eliminated with a stop order. It is documented here for transparency.

9. Capital Allocation and Throughput

- **Capital per trade:** 50% of total available capital
- **Concurrent positions:** Maximum 1 at any time
- **Cash floor:** Never less than 50% in cash
- **Trade frequency:** Approximately 1 trade per 1–3 trading days, depending on signal availability
- **Holding period:** Typically 1–3 trading days; up to 4 days including a weekend

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10. Risk Considerations

10.1 Inherent Strategy Risks

Risk	Description	Mitigation
Earnings Miss	Insiders can be wrong, or their purchases may reflect personal financial planning rather than earnings conviction	Stop-loss bracket; cash-first policy when signals are weak
Overnight/Weekend Gap	Adverse news can cause the stock to open well below the stop-loss trigger price	Accepted as inherent; not fully hedgeable without options
Thin Liquidity	Smaller-cap companies may have wide bid-ask spreads around earnings	Consider a minimum market cap filter (e.g., $500M+) — to be determined during backtesting
Signal Lag (Congress)	Congressional disclosures are delayed up to 45 days, meaning the market may have already moved	Algorithm uses transaction date, not disclosure date, for all window calculations
10b5-1 False Signals	Pre-scheduled purchase plans mimic insider conviction but are not real-time signals	Filtered out using the plan flag on Form 4 filings
Regulatory Change	SEC or STOCK Act disclosure rules could change	Monitor regulatory landscape; no active mitigation required at this time

10.2 What This Strategy Does Not Do

- **No short selling** — the algorithm is strictly long-only
- **No leverage** — positions are cash-funded only
- **No forced trading** — the algorithm holds cash when no qualifying candidate exists

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11. Open Items for Research

The following items were flagged during proposal development as needing further investigation before final implementation parameters are set. The architecture should be designed so these are configurable variables, not hardcoded values.

Item	Description	Recommended Action
Post-Earnings Exclusion Window	Exact number of days after prior earnings to exclude from insider signal consideration	Backtest at 10, 14, and 21 calendar days to determine which produces best predictive accuracy
Minimum Purchase Size	Should small symbolic purchases ($1,000–$5,000) count the same as large purchases?	Consider a minimum dollar threshold per purchase to filter noise; test multiple cutoffs
Market Cap Filter	Small-cap stocks may have liquidity issues that make execution difficult around earnings	Evaluate liquidity requirements; suggest testing a $500M minimum market cap floor
Stop-Loss Calibration	Is 4% the right stop-loss level for 1–3 day holds?	Analyze historical overnight volatility of mid-to-large cap stocks around earnings to calibrate
Score Threshold	What minimum composite score triggers a trade?	Determine via backtesting; target a threshold that balances signal quality with trade frequency
Late Window Multiplier	How much additional weight do Late Signal Window purchases deserve?	Backtest with 1.5x, 2x, and 2.5x multipliers
Sector Adjustments	Should insider signal weighting differ across sectors (e.g., tech vs. utilities)?	Evaluate in backtesting Phase 2 if initial results warrant refinement

12. Recommended Implementation Phases

Phase 1 — Data Infrastructure (Weeks 1–3)

- Build data pipelines for:
  - Yahoo Finance earnings calendar (via `yfinance`)
  - SEC EDGAR Form 4 filings (via `sec-edgar-downloader`)
  - House and Senate Stock Watcher APIs
- Normalize all data into a unified internal schema keyed by ticker and date
- Build the transaction-date extraction logic for congressional filings (handling the disclosure lag)

Phase 2 — Signal Engine Development (Weeks 4–6)

- Implement the timing window classifier
- Implement the composite scoring model (Section 6)
- Build the daily workflow logic (Section 7)
- Implement the ranking and selection logic (Section 9)

Phase 3 — Backtesting (Weeks 7–11)

- Backtest against at least 5 years of historical earnings events (2019–2024)
- Evaluate returns, win rate, average gain on winners, average loss on losers, and Sharpe ratio
- Use backtesting results to calibrate all open items from Section 11
- Ensure stop-loss logic and bracket order simulation are included in the backtest

Phase 4 — Paper Trading (Weeks 12–19)

- Run the algorithm in paper trading mode for a minimum of 2 months
- Validate live data pipeline reliability
- Confirm execution timing is achievable with the broker's order types
- Document edge cases (e.g., earnings date changes, trading halts, delistings)

Phase 5 — Live Deployment (Week 20+)

- Deploy with real capital following satisfactory paper trading results
- Monitor signal quality, execution quality, and P&L weekly

- Set a quarterly review cadence to re-evaluate scoring weights and thresholds

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13. Summary of Key Parameters

Parameter	Value	Notes
Capital per position	50%	Fixed; no exceptions
Max concurrent positions	1	Always
Minimum cash reserve	50%	Always
Stop-loss	4% below entry	Hard bracket order; to be calibrated
Entry timing	Near close on entry day	Final 15–30 min of session
Exit timing	Close on report processing day	Market-on-close order
Direction	Long only	No shorting
Trade forced when no signal	Never	Hold cash
Post-earnings exclusion window	14 calendar days	Adjustable; validate via backtest
Insider universe	C-suite, board, 10%+ owners, Congress	Form 4 + STOCK Act disclosures
10b5-1 plan trades	Excluded	Not genuine real-time signals
Data sources	All free	SEC EDGAR, Yahoo Finance, House/Senate Stock Watcher

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This proposal represents the initial design specification. All parameters marked as open items in Section 11 are intentionally left as configurable variables pending backtesting. The document should be revisited and updated following Phase 3 backtesting results.